**Aim:**

Write a C program to perform Merge sort. Display the partial pass-wise sorting done.

**Source Code:**

mergeSortAlgo.c

```
#include<stdio.h>
void display(int arr[20], int n);
void split(int arr[20], int low, int high);
void merge(int arr[20], int low, int mid, int high);
void display(int arr[20], int n){
   int i;
   for(i=0; i<n; i++){
      printf("%d ", arr[i]);
   }
   printf("\n");
}
void split(int arr[20], int low, int high){
   if(low<high){
      int mid = (low+high)/2;
      split(arr, low, mid);
      split(arr, mid+1, high);
      merge(arr,low,mid,high);
   }
}
void merge(int arr[20],int low, int mid, int high){
   int i=low,h=low,j=mid+1,k,temp[20];
   while(h<=mid && j<=high){
      if(arr[h]<=arr[j]){
         temp[i++]=arr[h++];
      }
      else{
         temp[i++]=arr[j++];
      }
   }
   while(h<=mid){
      temp[i++]=arr[h++];
   }
   while(j<=high){
      temp[i++]=arr[j++];
   }
   for(k=low;k<=high;k++){
      arr[k]=temp[k];
   }
   printf("Pass: ");
   for(k=low;k<=high;k++){
      printf("%d ",arr[k]);
   }
   printf("\n");
}
void main(){
   int arr[20],i,n;
```

```
    printf("no of elements: ");
    scanf("%d",&n);
    printf("elements: ");
    for(i=0;i<n;i++){
       scanf("%d",&arr[i]);
    }
    printf("Given array:\n");
    display(arr,n);
    split(arr,0,n-1);
    printf("Sorted array:\n");
    display(arr, n);
 }
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| User Output |
| no of elements: 5 |
| elements: 5 3 7 1 9 |
| Given array: |
| 5 3 7 1 9 |
| Pass: 3 5 |
| Pass: 3 5 7 |
| Pass: 1 9 |
| Pass: 1 3 5 7 9 |
| Sorted array: |
| 1 3 5 7 9 |

| Test Case - 2 |
|---|
| User Output |
| no of elements: 8 |
| elements: 8 4 2 7 1 5 3 6 |
| Given array: |
| 8 4 2 7 1 5 3 6 |
| Pass: 4 8 |
| Pass: 2 7 |
| Pass: 2 4 7 8 |
| Pass: 1 5 |
| Pass: 3 6 |
| Pass: 1 3 5 6 |
| Pass: 1 2 3 4 5 6 7 8 |
| Sorted array: |
| 1 2 3 4 5 6 7 8 |